Yifan Yu

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Education

University of Illinois Urbana-Champaign 2024 – Present
Ph.D. in Computer Science Urbana, IL

- Siebel School of Computing and Data Science

Georgia Institute of Technology 2020 – 2024
Bachelor of Science in Computer Science Atlanta, GA

- Concentration: Intelligence & Info Network

Research Interest

My research interests primarily focus on efficient machine learning, particularly efficient training and inference. Recently, I am interested in:

- Diffusion Language Modes; Multi-Agent Systems; Reinforcement Learning; Large Language Models; Inference Systems

Publications

6 first/cofirst-author accepted paper to ICML, ICLR, SOSP, COLM

*Equal contribution. †Equal advising. ⋄Corresponding author.

- **Introspective Diffusion Language Models** [arXiv] [Project page]
  Yifan Yu*, Yuqing Jian*, Junxiong Wang, Zhongzhu Zhou, Donglin Zhuang, Xinyu Fang, Sri Yanamandra, Xiaoxia Wu, Qingyang Wu, Shuaiwen Leon Song, Tri Dao, Ben Athiwaratkun, James Zou†, Fan Lai†⋄, Chenfeng Xu†⋄
  Conference on Language Modeling (COLM), 2026
- **CORRECT: COndensed eRror RECognition via knowledge Transfer in multi-agent systems** [arXiv]
  Yifan Yu, Moyan Li, Shaoyuan Xu, Jinmiao Fu, Xinhai Hou, Fan Lai, Bryan Wang
  International Conference on Machine Learning (ICML), 2026
- **XRPO: Pushing the Limits of GRPO with Targeted Exploration and Exploitation** [arXiv]
  Udbhav Bamba*, Minghao Fang*, **Yifan Yu***, Haizhong Zheng, Fan Lai
  International Conference on Machine Learning (ICML), 2026
- **IC-Cache: Efficient Large Language Model Serving via In-context Caching** [DOI]
  Yifan Yu*, Yu Gan*, Nikhil Sarda, Jiaming Shen, Lillian Tsai, Arvind Krishnamurthy, Yanqi Zhou, Henry M. Levy, David Culler, Fan Lai
  Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP), 2025
- **LoftQ: LoRA-Fine-Tuning-Aware Quantization for Large Language Models** [arXiv]
  Yixiao Li*, **Yifan Yu***, Chen Liang, Nikos Karampatziakis, Pengcheng He, Weizhu Chen, Tuo Zhao
  International Conference on Learning Representations (ICLR), 2024, **Oral Presentation**
- **LoSparse: Structured Compression of Large Language Models Based on Low-Rank and Sparse Approximation** [arXiv]
  Yixiao Li*, **Yifan Yu***, Qingru Zhang, Chen Liang, Pengcheng He, Weizhu Chen, Tuo Zhao
  International Conference on Machine Learning (ICML), 2023
- **Osprey: Target-agnostic Pre-training Makes Stronger Drafters in Speculative Decoding**
  Fengxiang Bie*, Yuqing Jian*, **Yifan Yu**, Zhongzhu Zhou, Zelei Shao, Ben Athiwaratkun, Shuaiwen Leon Song, Chenfeng Xu, Xiaoxia Wu, Tianyi Zhang
  The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)
- **OPPO: Accelerating PPO-based RLHF via Pipeline Overlap** [arXiv]
  Kaizhuo Yan, Yingjie Yu, **Yifan Yu**, Haizhong Zheng, Fan Lai
  International Conference on Learning Representations (ICLR), 2026

Preprints

- **Single-agent or Multi-agent Systems? Why Not Both?** [arXiv]
  Mingyan Gao*, Yanzi Li*, Banruo Liu*, **Yifan Yu**, Phillip Wang, Ching-Yu Lin, Fan Lai

arXiv preprint arXiv:2505.18286, 2025

Experience

Research Intern May 2026 – Now
Together AI Remote

- Working on efficient agentic diffusion language models with ultra-fast inference speed on edge devices.

Research Assistant 2024 – Present
University of Illinois Urbana-Champaign Urbana, IL

- Worked on efficient serving and post-training of LLMs/Multi-agent systems facing substantial numbers of requests, advised by Prof. Fan Lai.
- Implemented a system prototype that significantly improves the latency and throughput of LLM serving, as presented in the IC-Cache paper, by leveraging in-context learning theory; supported vLLM, Gemini, and LangChain backends.
- Implemented an efficient reinforcement learning framework via Verl that adaptively allocate the rollout budget with reward sharpening to improve RLVR efficiencies.
- Implemented an efficient online training frameworks for dllms

Research Intern May 2026 – Aug 2026
Together AI San Francisco, CA

- Working on next generation diffusion language models that achieve better pareto frontier of computation, efficiencies, and intelligence with online residual training.

Research Intern (Part-time) Jan 2026 – May 2026
Together AI Remote

- Worked on introspective diffusion language models, the first diffusion language model to achieve performance on par with autoregressive thinking models with introspective consistencies.
- Built training pipeline with LLaMA-Factory and SGLang-based inference, achieving 3.1–4× throughput gains over LLaDA-2.1 and SDAR series models.

Applied Scientist Intern May 2025 – Aug. 2025
Amazon Seattle, WA

- Worked on automatic error attribution for multi-agent systems.
- Proposed a skill engine for MAS error detection that significantly improves error detection accuracy; proposed synthesized datasets based on Magnetic One to support further research on automatic error attribution.

Student Researcher May 2024 – Jul. 2024
Systems Research@Google Remote / Seattle, WA

- Worked on efficient systems for serving LLMs facing millions of requests by leveraging the abundance of users' queries.
- Implemented experiments on millions of real users' requests using Hugging Face and vLLM backends; observed significant quality and throughput improvements on Alpaca, OpenOrca, and LMSYS-Chat.

Research Assistant Oct. 2021 – May 2024
Georgia Institute of Technology Atlanta, GA

- Worked on improving the efficiency of large language models with Prof. Tuo Zhao.
- Combined pruning methods and low-rank decomposition, as presented in the LoSparse paper; implemented the code and conducted experiments with Hugging Face and PyTorch.
- Proposed and implemented an alternating optimization method to minimize quantization error, as presented in the LoftQ paper; surpassed QLoRA across different precisions; adopted by PEFT and LLaMA-Factory.

Data Scientist Intern May 2022 – Jul. 2022
The Home Depot Atlanta, GA

- Worked in the online search team on DSI-based retrieval models as the first undergraduate intern in the team.
- Implemented DSI with PyTorch and Hugging Face; proposed an encoding method based on PECOS clustering for the DSI indexing process; improved recall by over 15% over the baseline, matching the current production search performance.

Service

- Reviewer: NeurIPS 2025, ICLR 2026, ICML 2026, COLM 2026, NeurIPS 2026.

Honors and Awards

- Presidential Undergraduate Research Award 2024

- Faculty Honor 2020 – 2023